Big data visual analytics systems

CS524: Big Data Visualization & Analytics

Fabio Miranda

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What is a visual analytics system?

What are the practical challenges in VA systems?

What are the research challenges in VA systems?

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What are the practical challenges in VA systems?

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Visual analytics

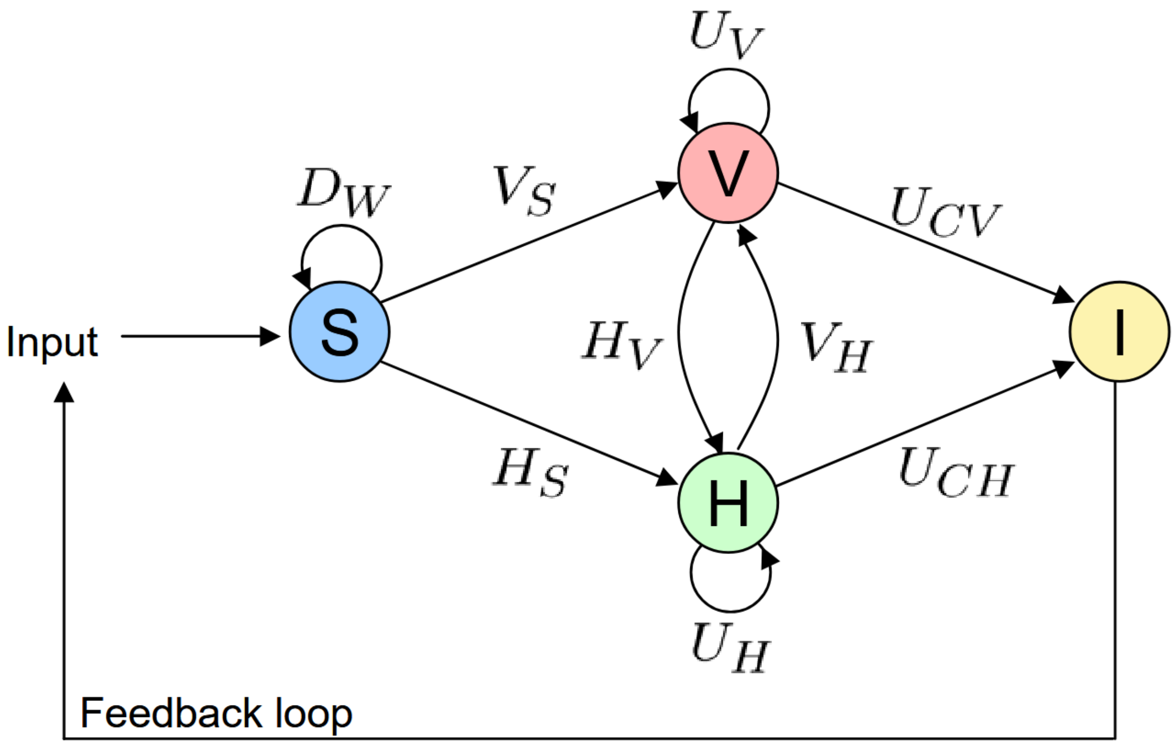
"Visual analytics is the formation of abstract visual metaphors in combination with a human information interaction that enables detection of the expected and discovery of the unexpected within massive, dynamically changing information spaces."

[Wong and Thomas, 2004]

Visual analytics, infovis, scivis

- **Scientific visualization:** data that has a natural geometric structure (wind flows, MRI).
- **Information visualization:** abstract representations (trees, graphs).
- **Visual analytics:** interactive visual representations <u>**and**</u> underlying analytical processes (ML, data mining, stats, etc.)

Visual analytics process



"Visual Analytics: Scope and Challenges", Keim et al., 2008

$F: S \rightarrow I$
$f \in \{D_w, V_x, H_y, U_z\}$
S: data, I: insight

D_w: data pre-processing functionalities (data transformation, data cleaning, data selection, data integration)

$V_x, x \in \{S, H\}$: visualization functions
- V_S: functions visualizing data ($S \rightarrow V$)
- V_H: functions visualizing hypotheses ($\mathrm{H} \rightarrow V$)

$H_y, y \in \{S, V\}$: hypotheses generation process
- H_S: hypotheses from automatic process ($S \rightarrow H$)
- H_v: hypotheses from visualization ($\mathrm{V} \rightarrow H$)

$U_z, z \in \{V, H, CV, CH\}$: user interactions
- U_V: visualization interaction ($V \rightarrow V$)
- U_H: hypotheses interaction ($\mathrm{H} \rightarrow H$)

U_{cv}: insight concluded from visualization
U_{ch}: insight concluded from hypothesis

Visual analytics process

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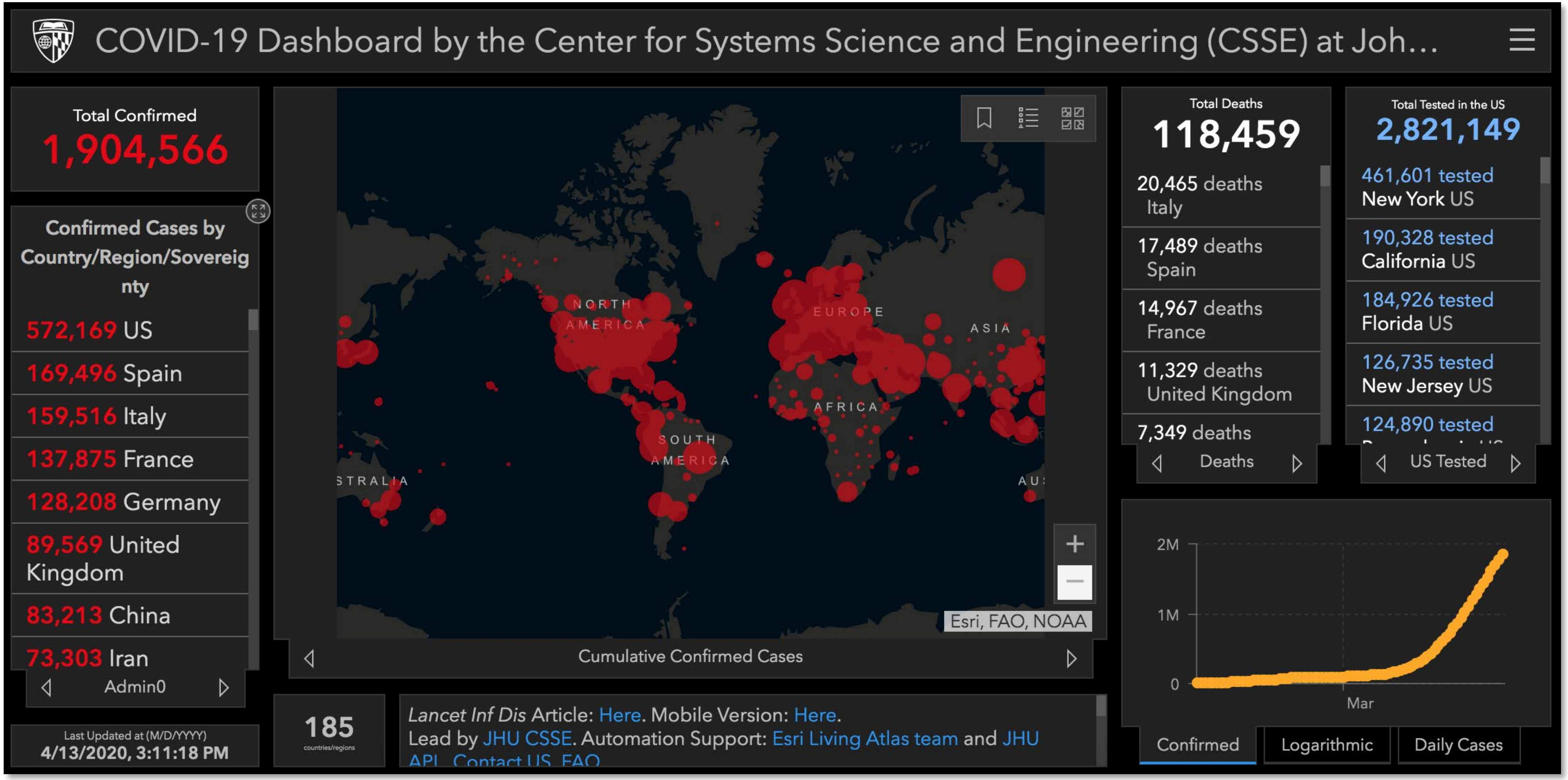
Spectrum of dashboards and VA systems



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Spectrum of dashboards and VA systems

Visual Analytics for the Big Data Era – A Comparative Review of State-of-the-Art Commercial Systems

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Abstract

Visual analytics (VA) system development started in academic research institutions where novel visualization techniques and open source toolkits were developed. Simultaneously, small software companies, sometimes spin-offs from academic research institutions, built solutions for specific application domains. In recent years we observed the following trend: some small VA companies grew exponentially; at the same time some big software vendors such as IBM and SAP started to acquire successful VA companies and integrated the acquired VA components into their existing frameworks. Generally the application domains of VA systems have broadened substantially. This phenomenon is driven by the generation of more and more data of high volume and complexity, which leads to an increasing demand for VA solutions from many application domains. In this paper we survey a selection of state-of-the-art commercial VA frameworks, complementary to an existing survey on open source VA tools. From the survey results we identify several improvement opportunities as future research directions.

Index Terms: H.4 [Information Systems]: INFORMATION SYSTEMS APPLICATIONS, K.1 [Computing Milieux]: THE COMPUTER INDUSTRY—Markets

1 Introduction

We are at the beginning of a big data era when data is generated at an incredible speed everywhere — from satellite images to social media posts, from online transaction records to high-throughput biological experiment results, and from mobile phone GPS signals to digital pictures and videos posted online [3]. According to IBM [9] 2.5 quintillion bytes of data are generated every day. Thus, 90% of todays data has been created in the last two years alone. This phenomenon leads to an increasing interest and effort from both academia and industry towards developing VA solutions with improved performance. On the academic side, a number of advanced VA techniques and open source toolkits have been developed [21]. On the industrial side, a large variety of companies, ranging from specialized data discovery vendors such as *Tableau*, *QlikTech*, and *TIBCO*, to multinational corporations such as *IBM*, *Microsoft*, *Oracle* and *SAP*, have all devoted much effort to develop their own commercial products for analyzing data of increasing volume and variety that arrives ever quicker.

Stakeholders from both academia and industry are well-aware of the importance of gaining an overview of the state-of-the-art solutions to stimulate innovative ideas and avoid redundant effort. Such overview enables people to understand limitations of existing solutions and thus to identify space for improvement. In the last couple of years, effort has been made to survey and compare the functionality of existing open-source VA toolkits [21] as well as commercial Business Intelligence (BI) applications [19, 28]. Such studies are important to assess what tools are available, what techniques they implement, and how good they are with respect to certain application tasks. However, a thorough survey of specific visual analysis functionality of existing commercial VA tools is still lacking, given that the range of tools in existing surveys is restricted to BI applications and focuses on the usability aspects of a product. Towards this end, we conducted a survey on a wider range of commercial VA tools including not only BI VA products but also a number of general purpose VA tools, and put our focus on evaluating their capability of handling data of large volume and variety efficiently. While existing surveys are largely based on user surveys, we devote much effort to evaluate the system performance and functionality by installing the software and testing with reference datasets.

We conducted our survey by first building an encompassing list of 15 relevant commercial systems. The choice is made by investigating current market share. A wide range of systems were selected, covering software that falls into different categories, for example, data discovery and visualization software, enterprise BI systems, network analysis toolkits, innovative and niche products; some products fall into more than one category. We assigned each system a priority level to make sure that we can focus on a smaller number of "core" systems without losing the whole picture. In the second phase, a structured questionnaire was designed for evaluating the functionality of each product from different perspectives, including *data management*, *visualization*, *automatic analysis*, and *system and performance*. We then contacted all vendors to get their answers to our questionnaire. Although many vendors responded with detailed answers, we did not manage to get responses from all of them.

In this paper we report the results for those ten systems whose vendors answered our questionnaire, including *Tableau* [14], *Spotfire* [4], *QlikView* [13], *JMP (SAS)* [11], *Jaspersoft* [10], *ADVIZOR Solutions* [6], *Board* [7], *Centrifuge* [8], *Visual Analytics* [15], and *Visual Mining* [16]. For the remaining systems in our initial list, some of which are regarded as key products in the market (*Cognos (IBM)*, *SQL Server BI (Microsoft)*, *Business Objects (SAP)*, *Teradata*, and *PowerPivot (Microsoft)*, we managed to find many answers to the questionnaire by ourselves, which allows us to gain a better understanding and overview of state-of-the-art VA systems.

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Four categories:

- **Exploration:** Allows users to generate and verify hypotheses, supports the ability to easily create and modify visualizations and statistical models.
- **Dashboard:** Fixed set of visualizations and controls with simple interactions (selection, filtering, drilling down).
- **Reporting:** Static summary of information from the data sources.
- **Alerting:** Automatic notification when the data sources reach predefined states.

Zhang et al., 2012

Spectrum of dashboards and VA systems

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A conceptual framework for developing dashboards for big mobility data

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ABSTRACT
Dashboards are an increasingly popular form of data visualization. Large, complex, and dynamic mobility data present a number of challenges in dashboard design. The overall aim for dashboard design is to improve information communication and decision making, though big mobility data in particular require considering privacy alongside size and complexity. Taking these issues into account, a gap remains between wrangling mobility data and developing meaningful dashboard output. Therefore, there is a need for a framework that bridges this gap to support the mobility dashboard development and design process. In this paper we outline a conceptual framework for mobility data dashboards that provides guidance for the development process while considering mobility data structure, volume, complexity, varied application contexts, and privacy constraints. We illustrate the proposed framework's components and process using example mobility dash-boards with varied inputs, end-users and objectives. Overall, the framework offers a basis for developers to understand how informational displays of big mobility data are determined by end-user needs as well as the types of data selection, transformation, and display available to particular mobility datasets.

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KEYWORDS
Dashboard; big mobility data; visualization; conceptual framework; design guidelines

KEY POLICY HIGHLIGHTS

- Defines essential components of big mobility dashboards for stakeholders to understand key considerations and data management/pre-processing needs
- Clarifies the differences between dashboards and visual analytics applications
- Provides guidance for gathering information from end-users to ensure displays are fit-for-purpose
- Illustrates the application of the conceptual framework for dashboard design by discussing several examples of mobility data dashboards

Introduction

Large volumes of individual mobility data have become available with the advent and prevalence of location aware technologies in recent years. While providing unprecedented detail about the ways that people and vehicles move through space, the nature of mobility data combined with their volume present a number of visualization challenges. Displaying massive mobility data quickly becomes visually overwhelming, thus making it difficult to understand patterns and behaviors and make informed decisions.

Dashboards have become a popular form of data visualization and interaction in a variety of domains including healthcare, community organizations, urban informatics, business, and education (Kitchin & McArdle, 2017; Li et al., 2020; Rivard & Cogswell, 2004; Sarikaya et al., 2018). The overall aim in deploying dashboards is to improve information comprehension and decision making, while synthesizing large or disparate data types and sources (Few, 2013). However, there is a gap between large and disparate data inputs and an effective dashboard output. Facing complex mobility data, developing a dashboard involves more than visualizing data with existing methods. Many additional aspects need to be considered, such as data type, volume, frequency of updates, the dashboard's purpose, and intended user base (Li et al., 2020; Stehle & Kitchin, 2020). Thus, bridging the gap between data inputs and usable dashboard outputs requires an analytical and systematic framework to guide dashboard development and design.

Dashboard design is data- and task-dependent; in light of the challenges and opportunities associated with mobility data, we set out to address issues related

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Conrow et al., 2022

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Both VA and dashboards facilitate analytical reasoning and decision-making by simplifying problems and communicating results visually.

However, they are different along three key aspects...

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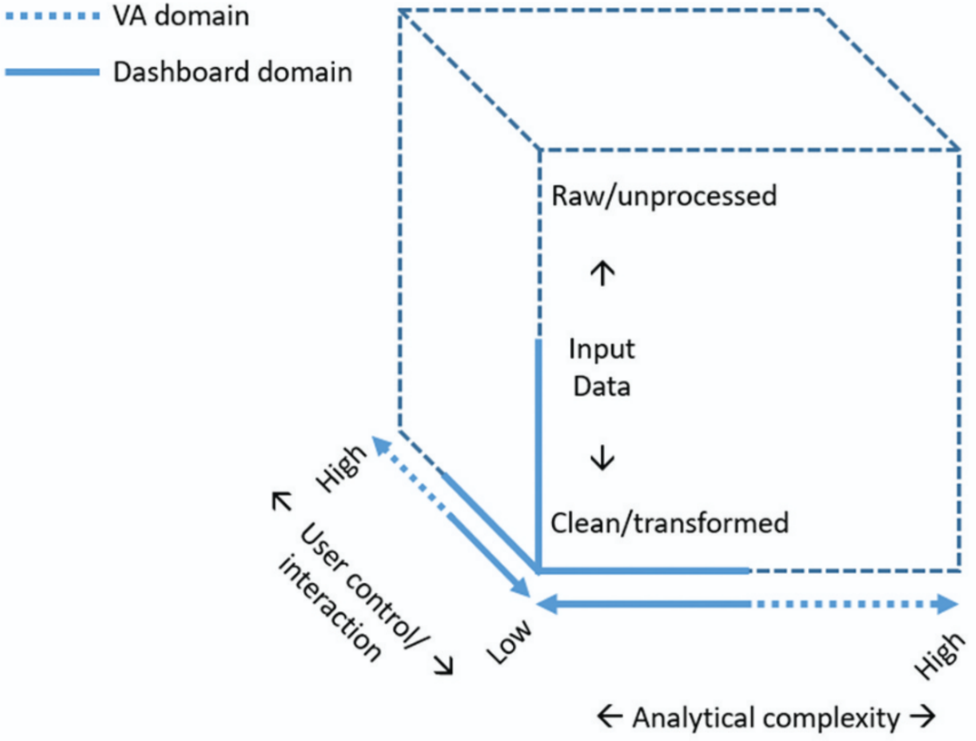

Figure 1. Comparison of visual analytics platforms and dashboards in the three key dimensions: input data, analytical complexity, and user interaction for output display.

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Spectrum of dashboards and VA systems

Input data

Dashboards: Data that are often already standardized (i.e., cleaned and integrated) and transformed (e.g., aggregated, re-aligned into a data cube) for display.

VA systems: Data handling and wrangling are part of the VA pipeline.

User interaction

Dashboards: At-a-glance views and monitoring for only relevant data points, focusing on a fixed aim, with standard tasks that are performed repeatedly.

VA systems: Enables higher flexibility as a user determines which processes and functions are appropriate.

Analytical complexity

Dashboards: Focus on informational displays rather than analyzing and manipulating input data, and just relying on simple selection, filtering, and comparison. Restricted to established objectives and displaying predefined knowledge outputs (i.e., results).

VA systems: Incorporates analytical processes (e.g., ML, data mining) that produce insights from large, complex data through an iterative or looping process. Data analysis as a primary and iterative process for knowledge discovery and hypothesis testing.

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Practical challenges in big data VA systems

- Unstructured and heterogeneous data
- Advanced visualizations and interactions
- Customizable visualizations
- Real-time and interactive queries
- Analytics integration
- Multi-resolution reasoning

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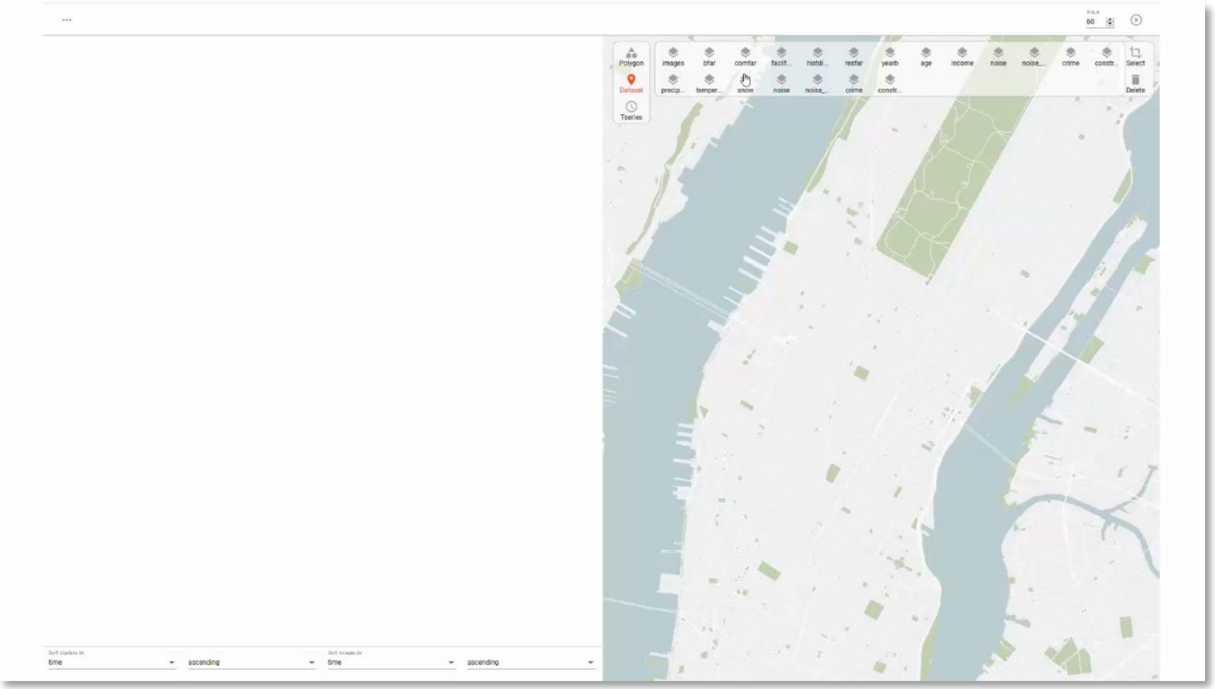
Miranda et al., 2020



Big data VA systems may need to handle text, images, sensors, etc. with inconsistent schemas and identifiers. How to align data so that they can be joined across sources?

Practical challenges in big data VA systems

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- **Advanced visualizations and interactions**
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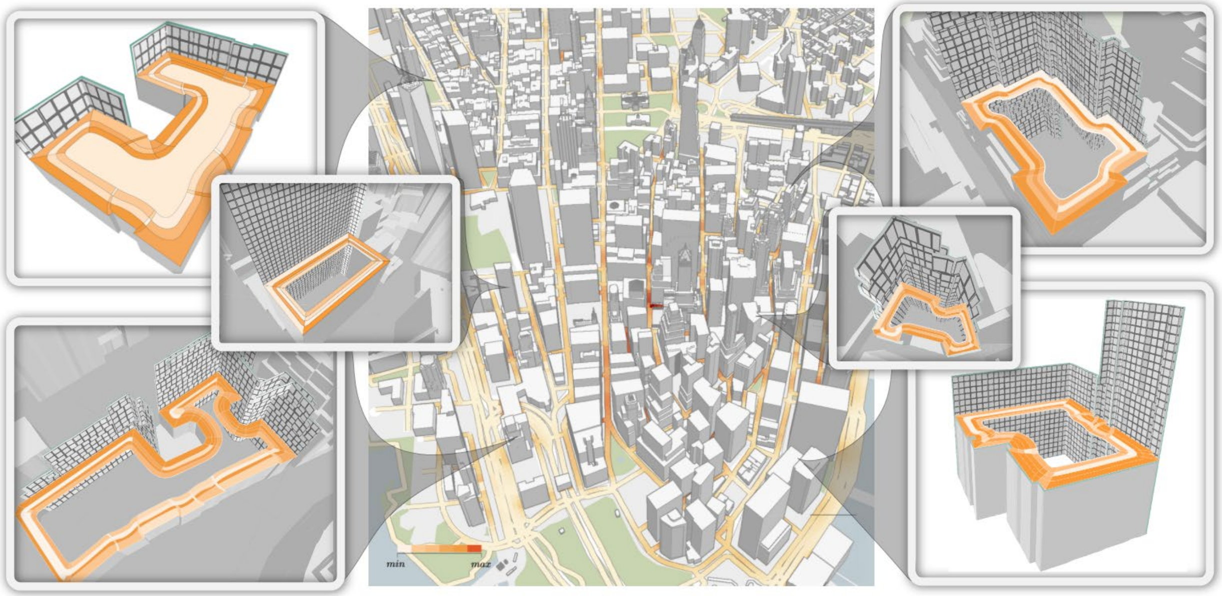
Mota et al., 2026



Innovative visualization and interaction techniques may be needed to handle complex data, beyond "standard" techniques such as line charts, bar charts, and scatter plots. What is the design space of such visualizations?

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Practical challenges in big data VA systems

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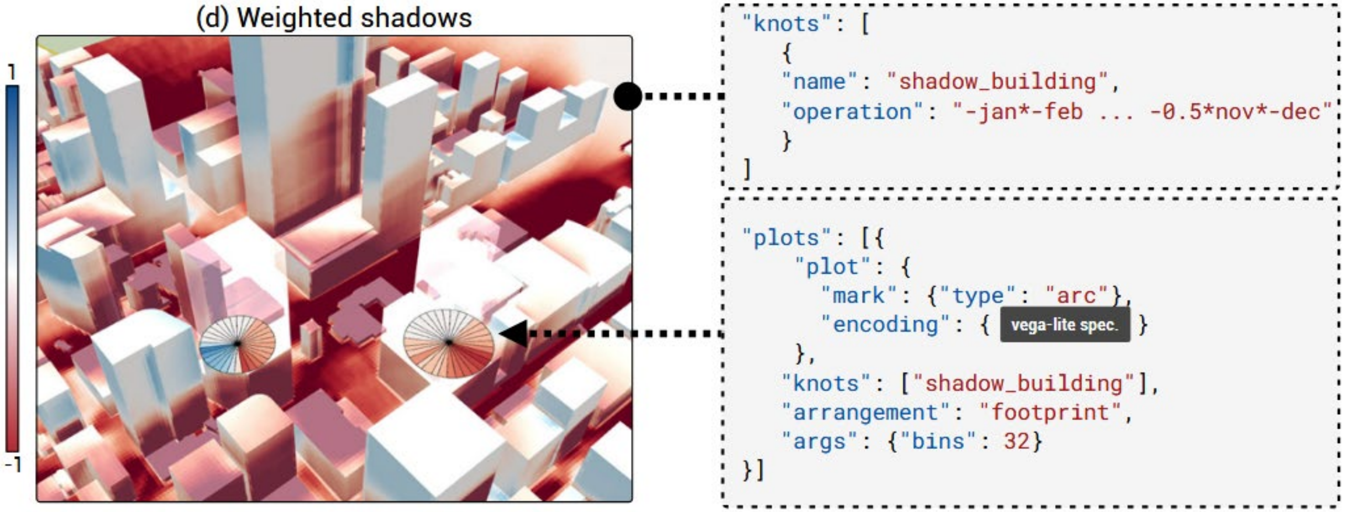
Moreira et al., 2023



Given the same data and technique, different parameters may lead to different representations and impressions. User must be able to iterate over the design space to gain insights from the visualization. How to do so while handling complex and large data in a complex design space?

Practical challenges in big data VA systems

- Unstructured and heterogeneous data
- Advanced visualizations and interactions
- Customizable visualizations
- **Real-time and interactive queries**
- Analytics integration
- Multi-resolution reasoning

Wang et al., 2021

VA systems must support interactivity; users need sub-second feedback to explore hypotheses and maintain cognitive flow. How to support such queries? How to communicate partial results and uncertainty?

Practical challenges in big data VA systems

- Unstructured and heterogeneous data
- Advanced visualizations and interactions
- Customizable visualizations
- Real-time and interactive queries
- **Analytics integration**
- Multi-resolution reasoning

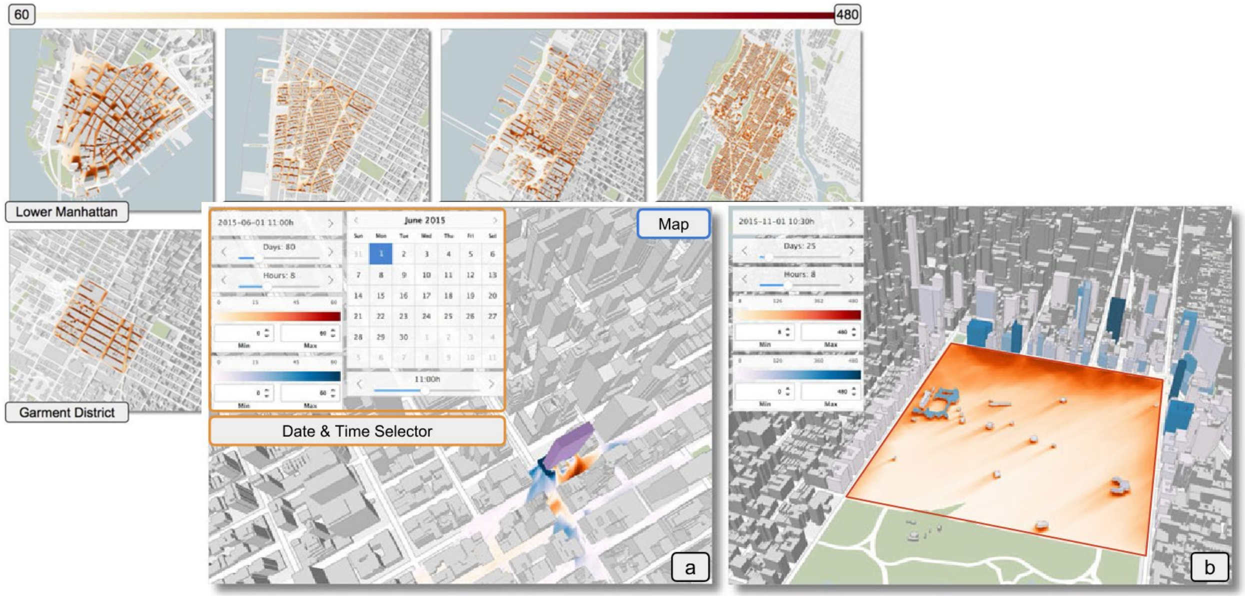
Doraiswamy et al., 2015

The problem might require models and ML embedded into the same workflow. Integration is not straightforward as VA system introduces parameters, assumptions, and failure modes that must be controlled. How to integrate analytics while supporting provenance and explainable controls?

Practical challenges in big data VA systems

- Unstructured and heterogeneous data
- Advanced visualizations and interactions
- Customizable visualizations
- Real-time and interactive queries
- Analytics integration
- **Multi-resolution reasoning**

Miranda et al., 2019



Big data might requires jumping between overview patterns and local explanations (e.g., city → neighborhood → street; year → day → minute). patterns may appear or disappear depending on binning, boundaries, and sampling. How to support interaction across resolutions consistently?

Practical challenges in big data VA systems

- Unstructured and heterogeneous data
- Advanced visualizations and interactions
- Customizable visualizations
- Real-time and interactive queries
- Analytics integration
- Multi-resolution reasoning

In practice, a big data VA system might need to tackle multiple (if not all!) of these challenges!

Visual analytics + big data

“We propose a new model that allows users to visually query taxi trips. Besides standard analytics queries, the model supports origin-destination queries that enable the study of mobility across the city. We show that this model is able to express a wide range of spatiotemporal queries, and it is also flexible in that not only can queries be composed but also different aggregations and visual representations can be applied, allowing users to explore and compare results.”

Visual Exploration of Big Spatio-Temporal Urban Data: A Study of New York City Taxi Trips

Nivan Ferreira, Jorge Poco, Huy T. Vo, Juliana Freire, and Cláudio T. Silva

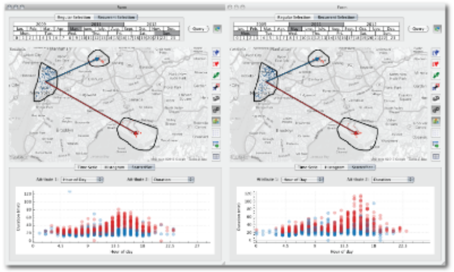


Fig. 1. Comparison of taxi trips from Lower Manhattan to JFK and LGA airports in May 2011. The query on the left selects trips that occurred on Sundays, while the one on the right selects trips that occurred on Mondays. Users specify these queries by visually selecting regions on the map and connecting them. In addition to inspecting the results depicted on the map, i.e., the dots corresponding to pickups (blue) and dropoffs (orange) of the selected trips, they can also explore the results through other visual representations. The scatter plots below the maps show the relationship between hour of the day and trip duration. Points in the plots are colored according to the spatial constraint represented by the arrows between the regions: trips to JFK in blue, and trips to LGA in red. The plots show that many of the trips on Monday between 3PM and 5PM take much longer than trips on Sundays.

Abstract—As increasing volumes of urban data are captured and become available, new opportunities arise for data-driven analysis that can lead to improvements in the lives of citizens through evidence-based decision making and policies. In this paper, we focus on a particularly important urban data set: taxi trips. Taxis are valuable sensors and information associated with taxi trips can provide unprecedented insight into many different aspects of city life, from economic activity and human behavior to mobility patterns. But analyzing these data presents many challenges. The data are complex, containing geographical and temporal components in addition to multiple variables associated with each trip. Consequently, it is hard to specify exploratory queries and to perform comparative analyses (e.g., compare different regions over time). This problem is compounded due to the size of the data—there are on average 500,000 taxi trips each day in NYC. We propose a new model that allows users to visually query taxi trips. Besides standard analytics queries, the model supports origin-destination queries that enable the study of mobility across the city. We show that this model is able to express a wide range of spatio-temporal queries, and it is also flexible in that not only can queries be composed but also different aggregations and visual representations can be applied, allowing users to explore and compare results. We have built a scalable system that implements this model which supports interactive response times; makes use of an adaptive level-of-detail rendering strategy to generate clutter-free visualization for large results; and shows hidden details to the users in a summary through the use of overlay heat maps. We present a series of case studies motivated by traffic engineers and economists that show how our model and system enable domain experts to perform tasks that were previously unattainable for them.

Index Terms—Spatio-temporal queries; urban data; taxi movement data; visual exploration

1 INTRODUCTION

For the first time in history, more than half of the world's population lives in urban areas. Enabling cities to deliver services effectively, efficiently, and sustainably is among the most important undertakings in this century. While in the recent past, decision makers and social scientists faced significant constraints in obtaining the data needed to understand city dynamics and evaluate policies and practices, data are now abundant. Many cities have started to make a wide range of data sets available, see e.g., [24, 10, 7]. The challenge now is how to make sense of these data.

We examine one particularly important urban data set: taxi trips. In New York City, each day 13,000 taxis carry over one million passengers and make, on average, 500,000 trips—totaling over 170 million trips a year. Taxi trips are thus valuable sensors of city life. Consider the plot in Fig. 2, which shows how the number of trips per day varies

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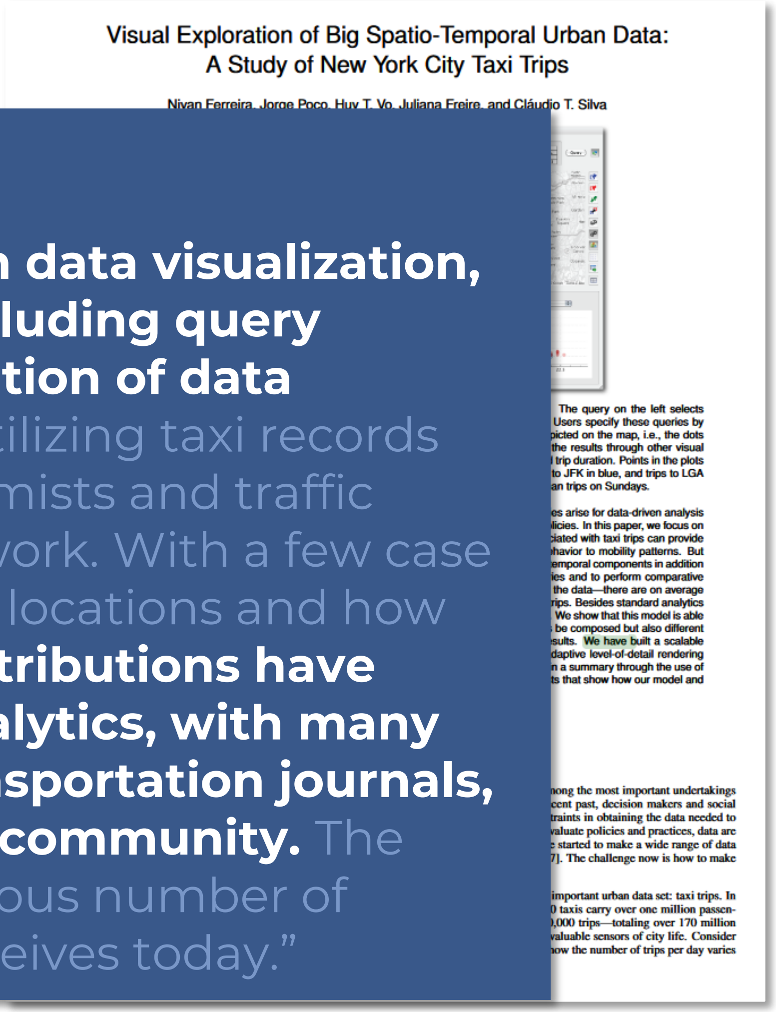
Manuscript received 31 March 2013; accepted 1 August 2013; posted online 13 October 2013; mailed on 4 October 2013. For information on obtaining reprints of this article, please send e-mail to: tvcg@computer.org.

[Ferreira et al., 2013]

Visual analytics + big data

“We propose a new model that allows users to vis… analy… origin… study… that t… range… also f… comp… and v… allow… result…


Visual Exploration of Big Spatio-Temporal Urban Data:
A Study of New York City Taxi Trips

Nivan Ferreira, Jorge Poco, Huy T. Vo, Juliana Freire, and Cláudio T. Silva

Test-of-time award in 2023

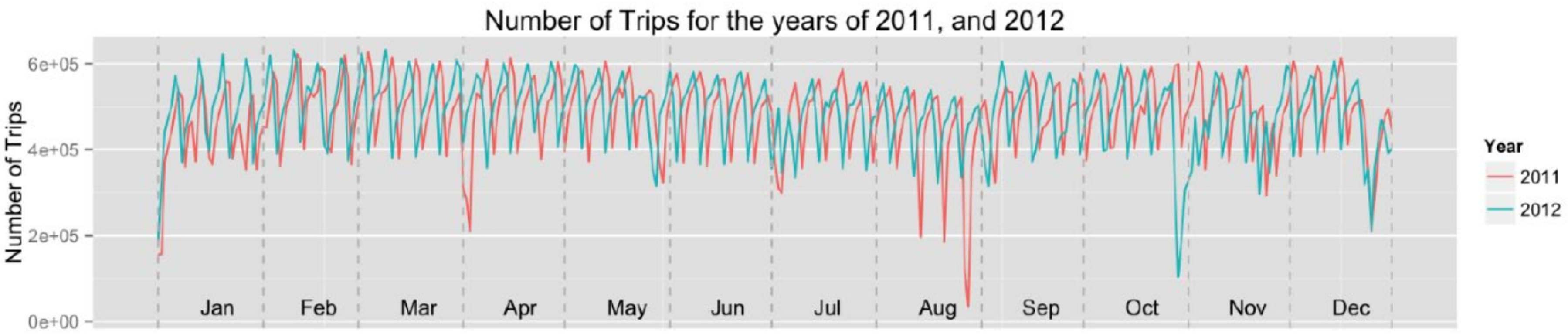
“**This paper is one of the landmark papers in urban data visualization, making several truly impressive contributions, including query language developments and the seamless integration of data querying to handle real-world, large-scale data.** Utilizing taxi records from New York City, the authors engage with economists and traffic engineers to demonstrate the power of their framework. With a few case studies, they explore the economic incentives of taxi locations and how traffic patterns impact demand patterns. **These contributions have situated this paper as a seminal work in urban analytics, with many citations in visualization, urban planning, and transportation journals, making a clear impact within and outside the VIS community.** The impact of this work is also expressed in the tremendous number of citations this paper has received thus far and still receives today.”

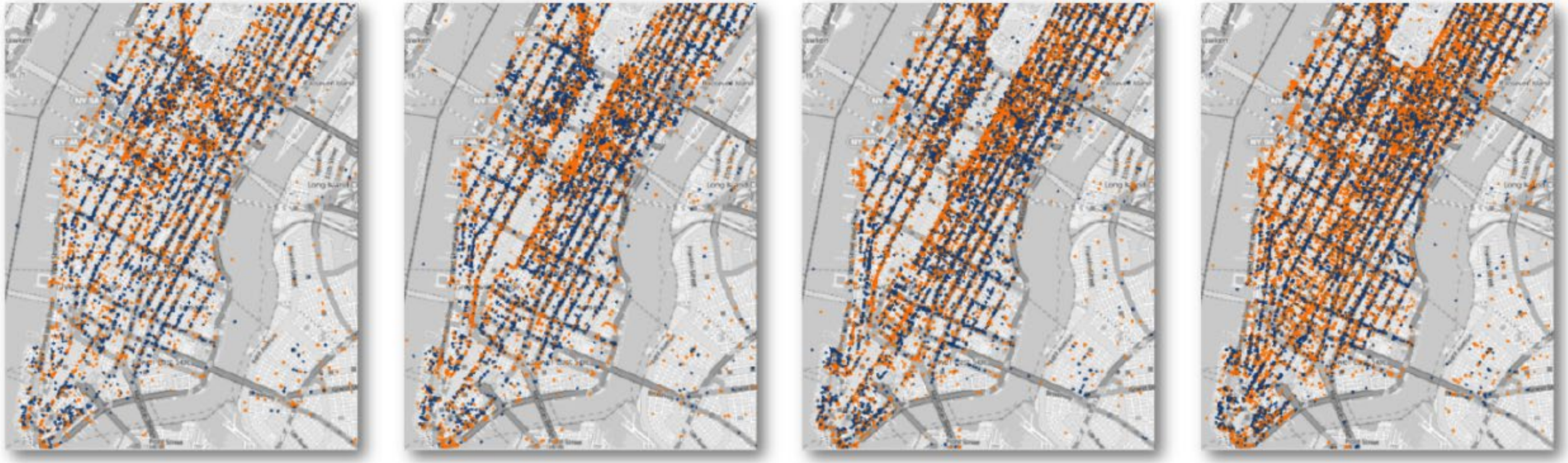
[Ferreira et al., 2013]

Taxi as sensors of city life

- Understanding the dynamics of the city, how different aspects of the data vary over space and time:
  - “*What is the average trip time from Midtown to the airports during weekdays?*”
  - “*How does the movement changes between Midtown and JFK throughout the day, over different days of the week?*”
  - “*How does the taxi fleet activity vary during weekdays?*”
- Ability to quickly test hypotheses: starting with one query about a specific place (“*movement patterns between Midtown and JFK*”) and generalize to all neighborhoods.

Taxi as sensors of city life



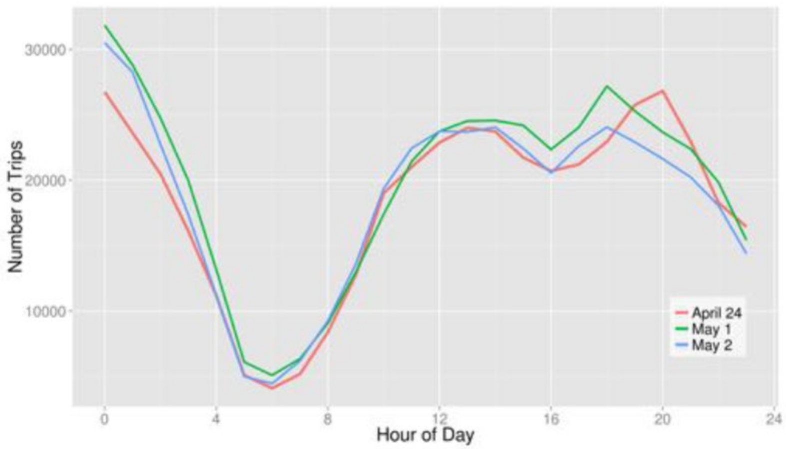


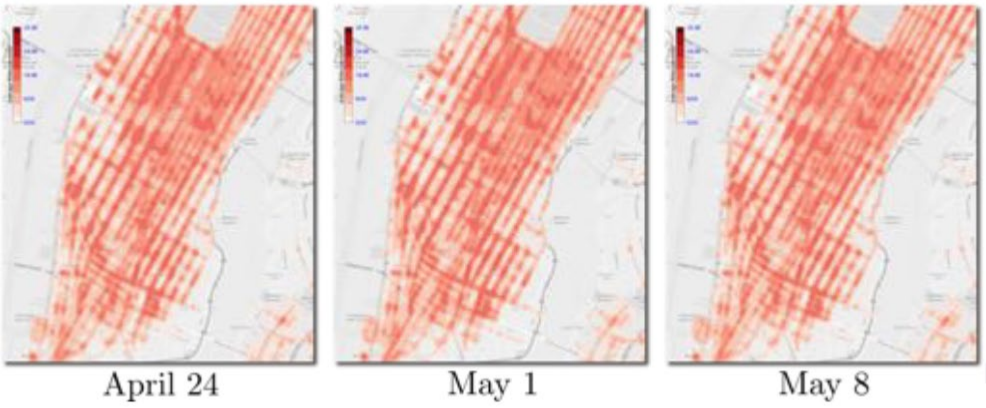
Visual analysis process: taxi data

- Data: ~500,000 trips / day; ~1,000,000,000 trips in 5 years
  - Spatiotemporal: pickup + dropoff
  - Trip attributes: distance traveled, fare, tip, etc.
- Government, policy makers and scientists are usually unable to *interactive* explore the *whole* data.
  - Too many slices to examine.

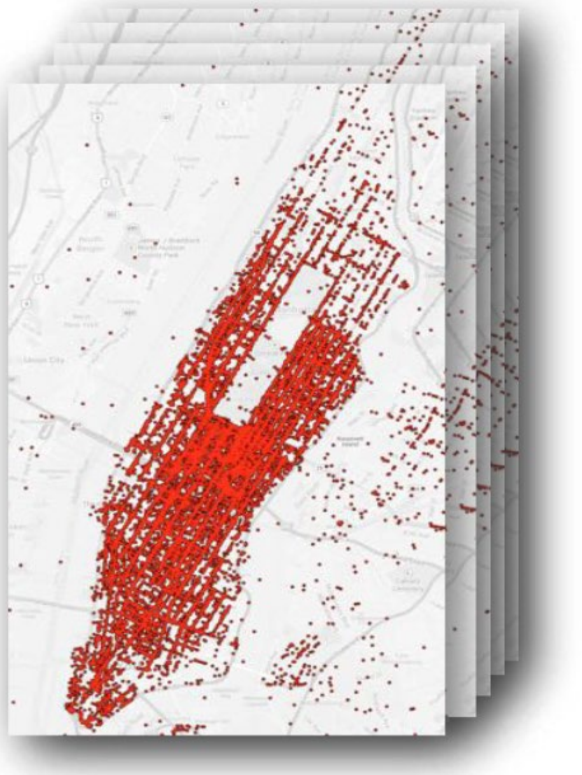
Visual analysis process: taxi data

- Too many slices
- 365*24 1-hour slices in just one year
- Which slices are interesting?





Visual analysis process: taxi data



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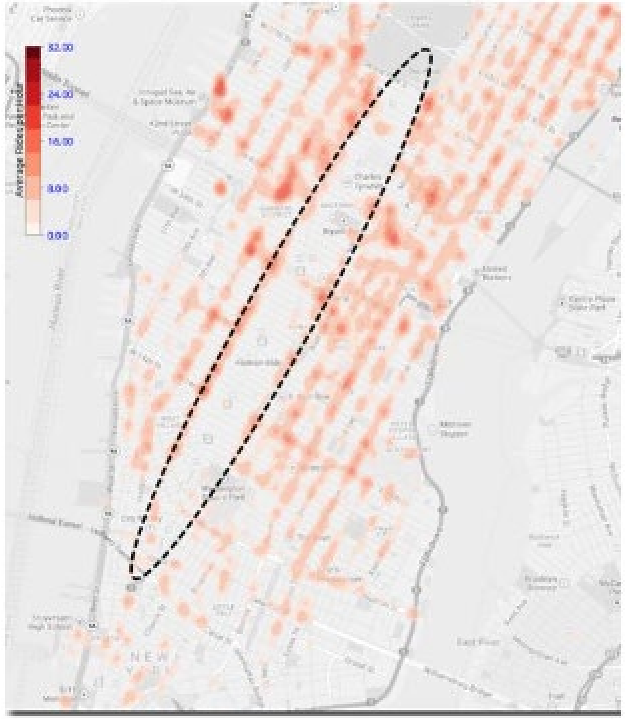

May 1 (8-9am)

Visual analysis process: taxi data

- Too many slices
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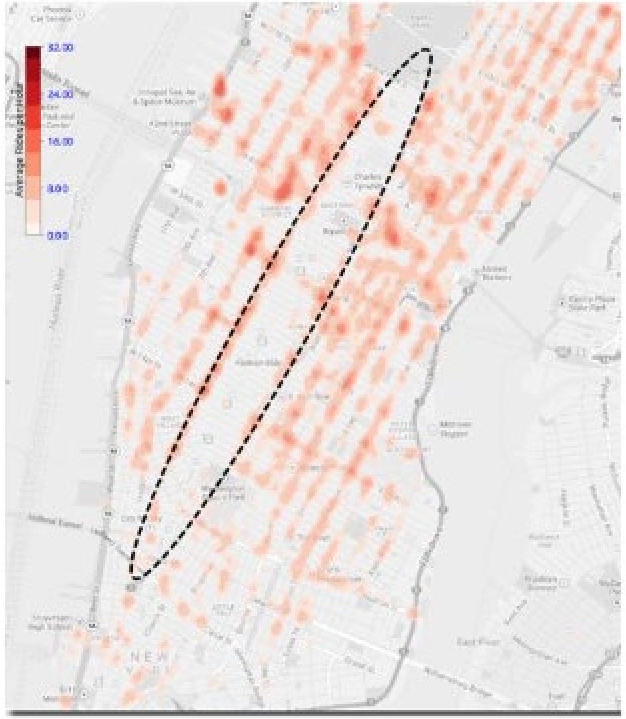

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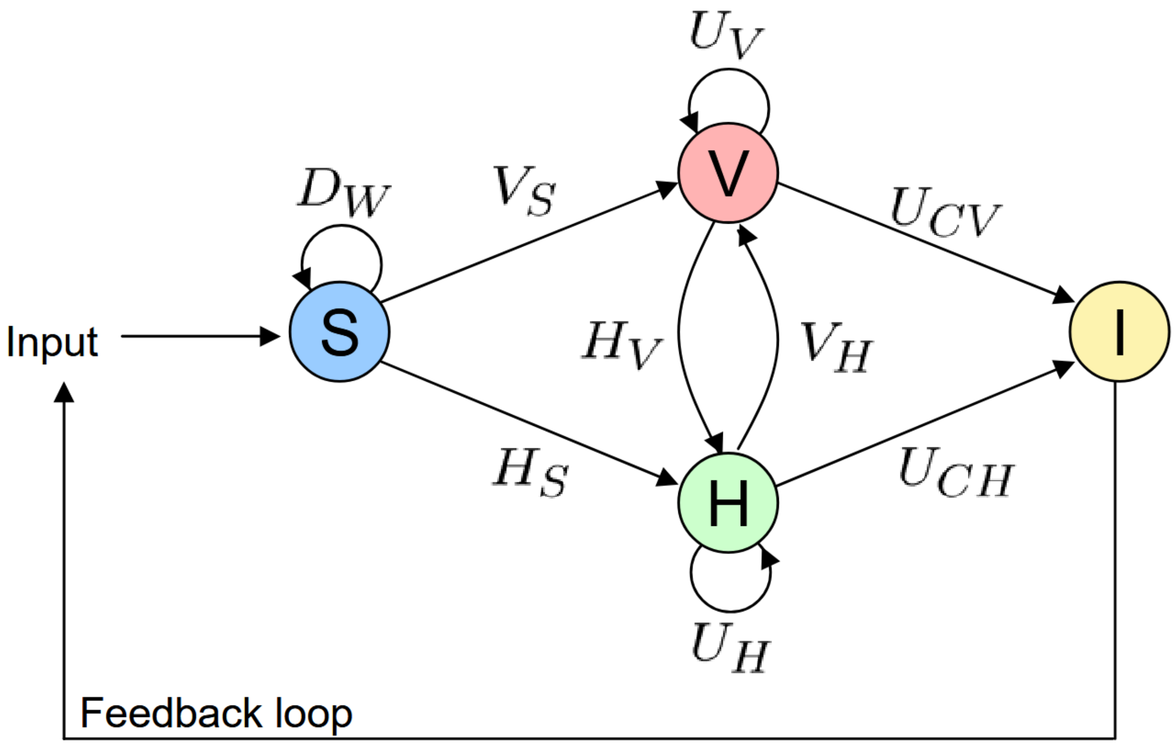
Guide users towards *interesting* data slices

Visual analytics process



"Visual Analytics: Scope and Challenges", Keim et al., 2008

$F: S \rightarrow I$
$f \in \{D_w, V_x, H_y, U_z\}$
S: data, I: insight

D_w: data pre-processing functionalities (data transformation, data cleaning, data selection, data integration)

$V_x, x \in \{S, H\}$: visualization functions
- V_S: functions visualizing data ($S \rightarrow V$)
- V_H: functions visualizing hypotheses ($\mathrm{H} \rightarrow V$)

$H_y, y \in \{S, V\}$: hypotheses generation process
- H_S: hypotheses from automatic process ($S \rightarrow H$)
- H_v: hypotheses from visualization ($\mathrm{V} \rightarrow H$)

$U_z, z \in \{V, H, CV, CH\}$: user interactions
- U_V: visualization interaction ($V \rightarrow V$)
- U_H: hypotheses interaction ($\mathrm{H} \rightarrow H$)

U_{cv}: insight concluded from visualization
U_{ch}: insight concluded from hypothesis

Visual analysis process: taxi data

- Too many slices
- 365*24 1-hour slices in just one year
- Which slices are interesting?

Guide users towards *interesting* data slices

Taxi pickups and drop-offs: data sets $S_1, S_2, \ldots, S_n$

Pre-processing:

- Data cleaning (e.g., remove points outside NYC): $d_c(S_1, \ldots, S_n)$
- Data integration (e.g., add weather column): $d_i(S_1, \ldots, S_n)$
- Data transformation (e.g., raw csv's to binary): $d_t(S_1, \ldots, S_n)$
- Data selection (e.g., only Manhattan trips): $d_s(S_1, \ldots, S_n)$

Visual analysis process: taxi data

- Too many slices
- 365*24 1-hour slices in just one year
- Which slices are interesting?

Guide users towards *interesting* data slices

Data mining:

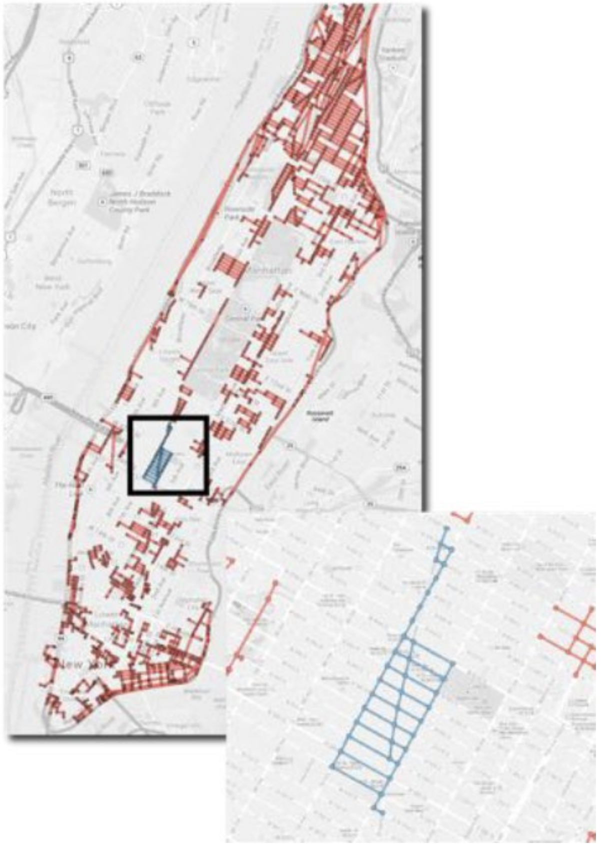
Identifying events with arbitrary spatial structure at multiple temporal scales:

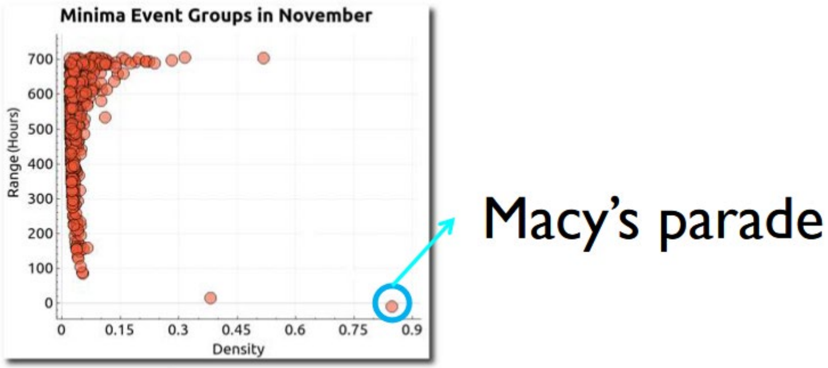
$$h' = h_s(S')$$
$$\text{with } S' = d_s(d_t(d_i(d_c(S_1, \dots, S_n))))$$

Visual analysis process: taxi data

Visualization:

- Visualizing the previous hypothesis: $v' = v_h(h')$
- Adjusting parameters: $v'' = u_v(v')$
- Hypotheses from visualization: $h'' = h_v(v'')$, leading to an insight (unusual taxi activity around Macy's parade).





Visual analysis process: taxi data

- Taxi pickups and drop-offs: data sets $S_1, S_2, \dots, S_n$
- Pre-processing
  - Data cleaning (e.g., removing): $d_c(S_1, \dots, S_n)$
  - Data integration (e.g., adding extra attributes): $d_i(S_1, \dots, S_n)$
  - Data transformation (e.g., format suitable for analysis): $d_t(S_1, \dots, S_n)$
  - Data selection (e.g., subset of sensors): $d_s(S_1, \dots, S_n)$
- Data mining:
  - Identifying events with arbitrary spatial structure at multiple temporal scales:

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Visual analysis process: taxi data

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$$h_v(u_v(v_h(h_s(d_s(d_t(d_i(d_c(S_1, \dots, S_n))))))))$$

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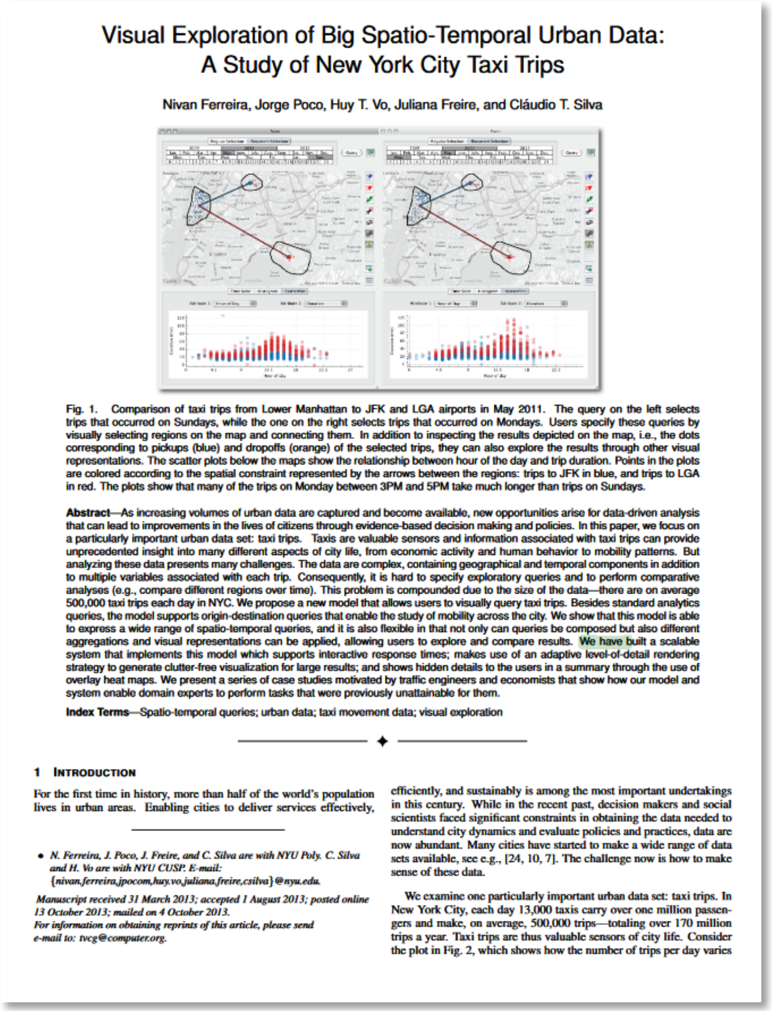
Visual analytics + big data

- Interaction capabilities that enable users to pose queries over all the dimensions of the data and flexibly explore the associated attributes.
- Compare spatiotemporal slices through multiple coordinated views.
- Interactively compose and refine queries, and generalize them by performing parameter sweeps.
- Efficient data stored, and adaptive level-of-detail rendering to provide clutter-free visualizations.



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Nivan Ferreira, Jorge Poco, Huy T. Vo, Juliana Freire, and Cláudio T. Silva

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[Ferreira et al., 2013]

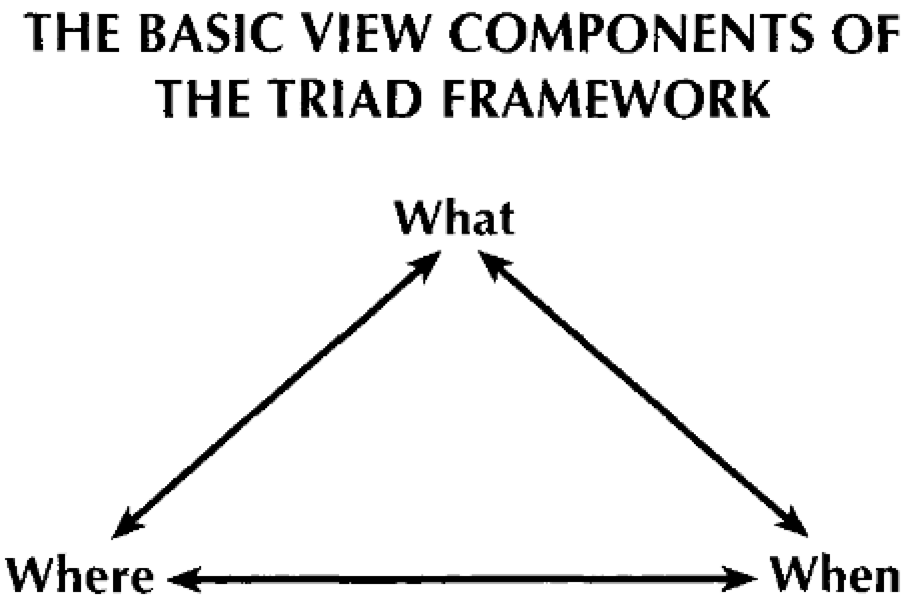
Usability through visual operations (u)

```
SELECT *
FROM trips
WHERE pickup_time in (5/1/11,5/7/11)
AND dropoff_loc in “Times Square”
AND pickup_loc in “Gramercy”
```

Data selection and result exploration are unified

Visual query model (u)

- Expressive Triad framework:
  - *Where + when → what*: describe the objects (what) that are present at certain locations and times.
  - *When + what → where*: describe the locations occupied by objects at a given time.
  - *Where + what → when*: describe the times that objects occupied a given location.



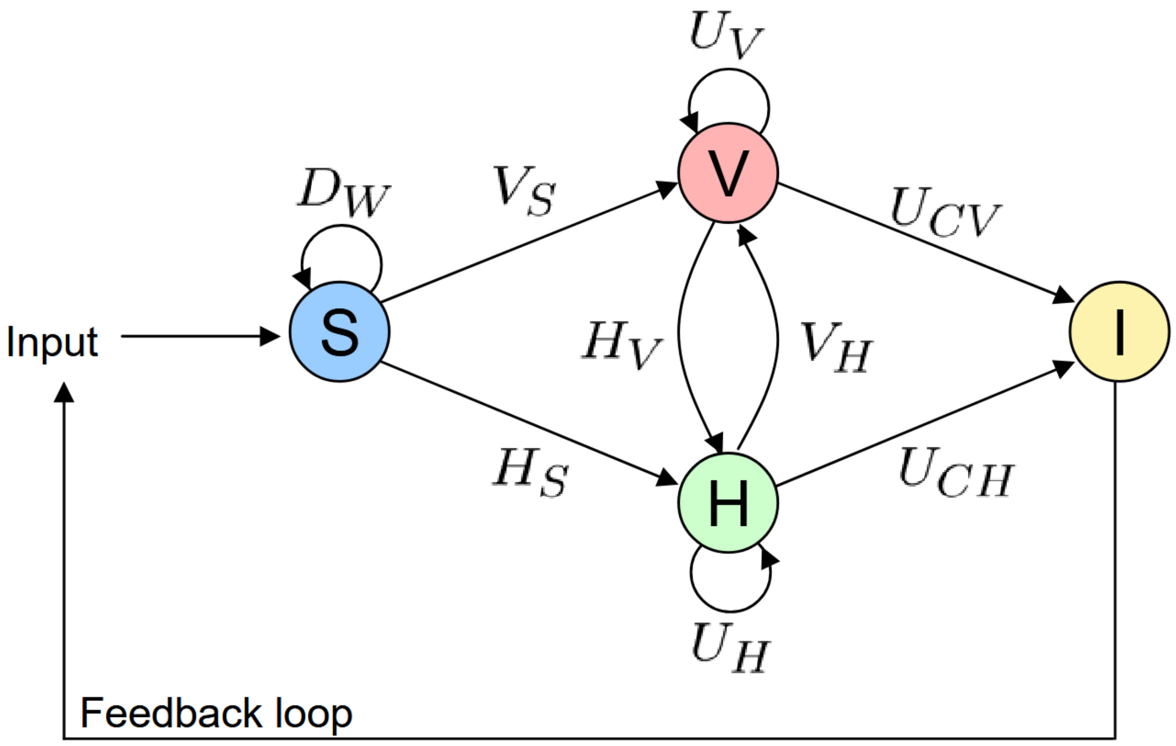
[Peuquet, 1994]

Visual query model (u)

- Expressive Triad framework:
  - *Where + when → what*: “What is the average trip time from Midtown to the airports during weekdays?”
  - *When + what → where*: “Where are the hot spots in Manhattan in the weekends?”
  - *Where + what → when*: “When were activities restored in Lower Manhattan after the Sandy hurricane?”

What? Where? When?

Finding interesting features (h_s)



How to tackle h_s (hypotheses from automatic methods)?

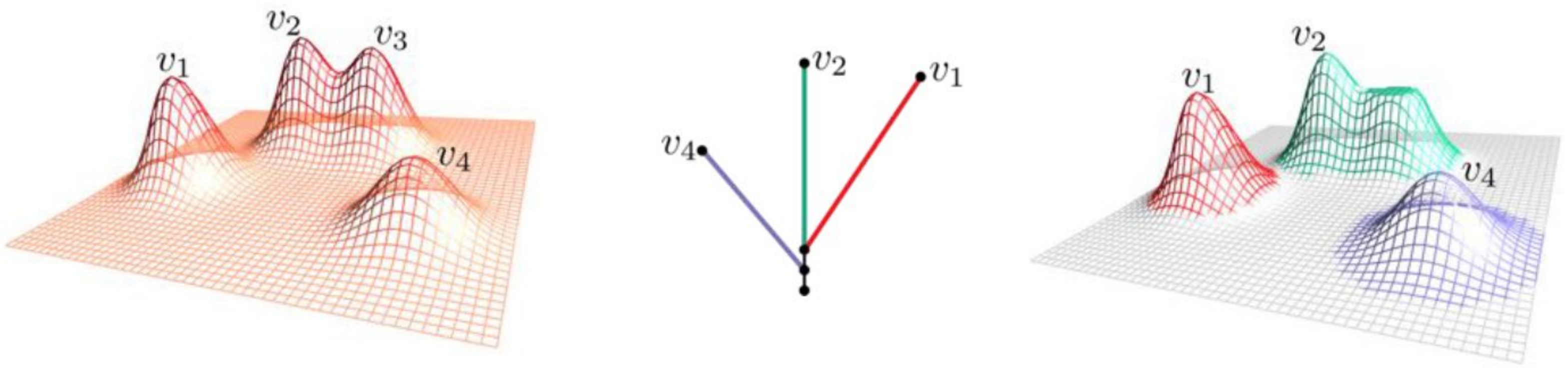
Goal: guide users towards *interesting* data slices.

Finding interesting features (h_s)

- Model data as a time-varying scalar function defined on a graph.
  - Taxi data: graph = road network; function = density of taxis
  - Subway data: graph = track network; function = delay of trains

Finding interesting features (h_s)

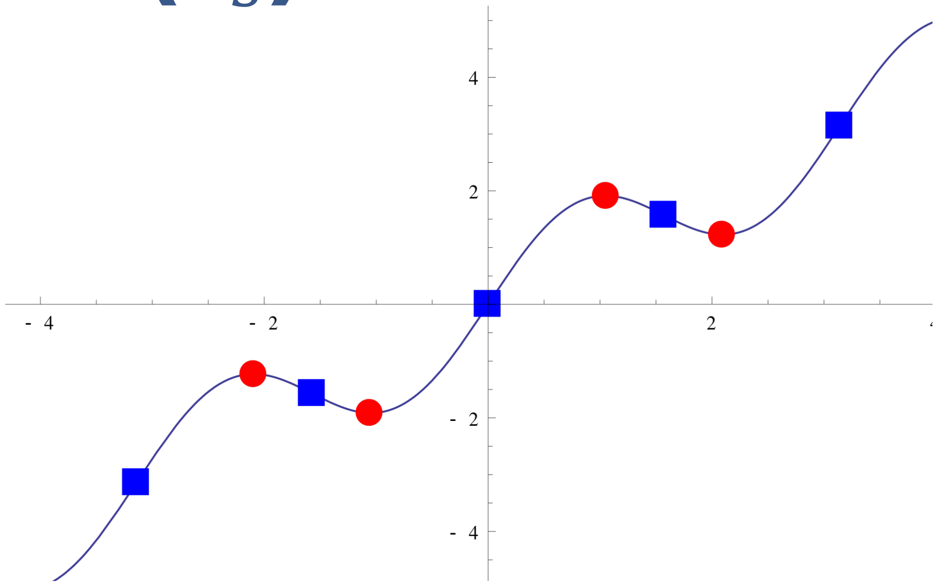
- Merge trees to efficiently identify events in each time step.
- Set of potential events: regions corresponding to maxima and minima.
  - A region is interesting if its behavior differs from its neighborhood.
  - Unimportant events can be simplified.

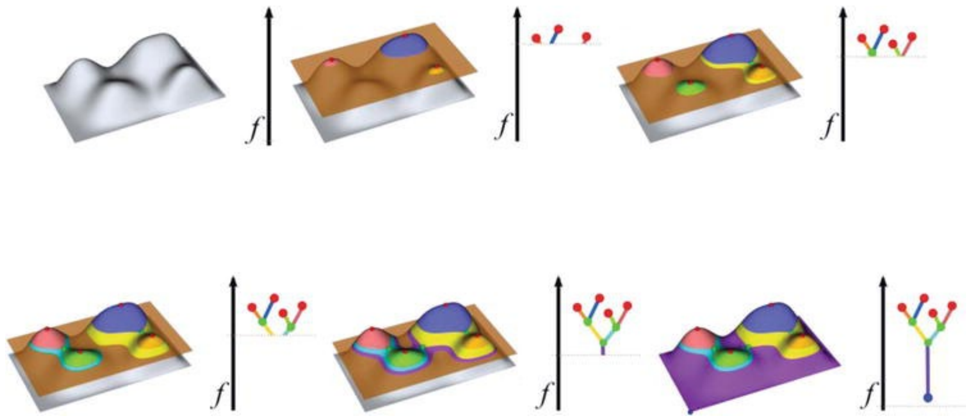


[Doraiswamy et al., 2014]

Finding interesting features (h_s)

- Merge trees can be used to efficiently represent regions
- Topological changes occur at critical points (i.e., where the function is not differentiable or derivative equal to zero).
- Trees can be simplified to remove noise.
- Merge trees track level sets and capture local maxima as function height is reduced.

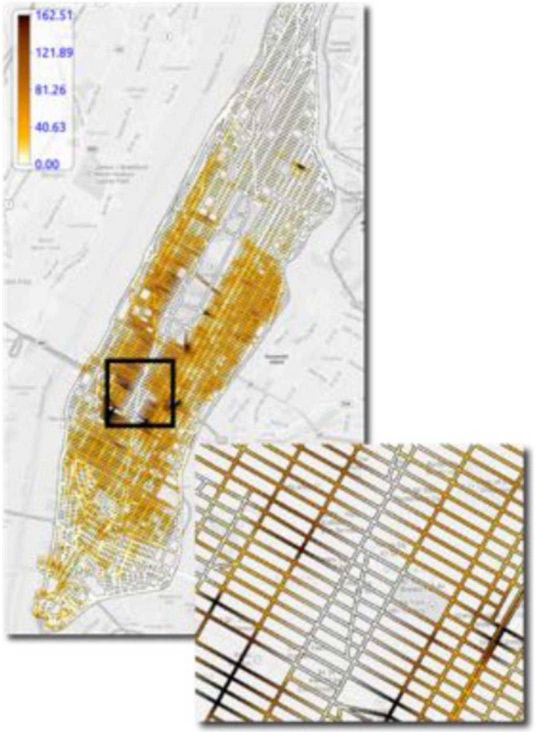




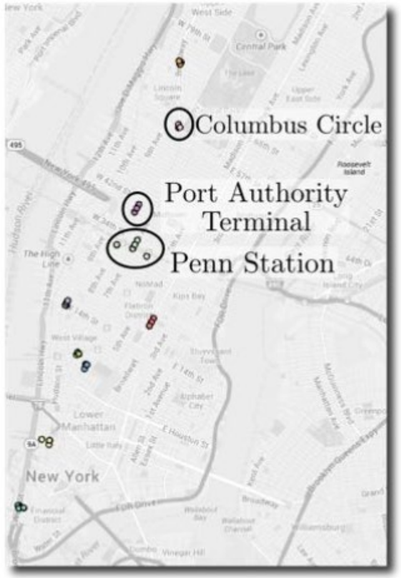
[Thomas et al., 2018]

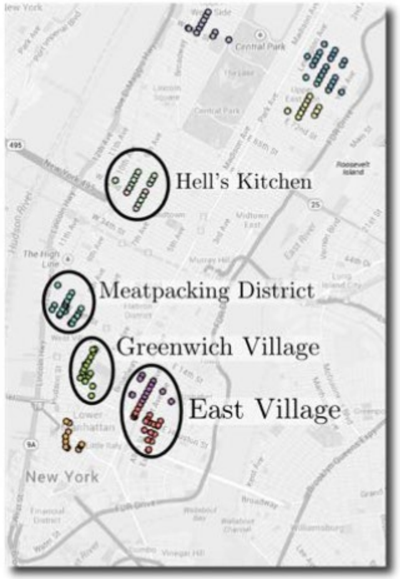
Finding interesting features (h_s)

- Minima: lack of taxis
  - Density lower than local neighborhood.
  - Can denote road blocks.



Maxima: popular taxi locations
  - Density higher than local neighborhood.
  - Can denote tourist locations, train stations.





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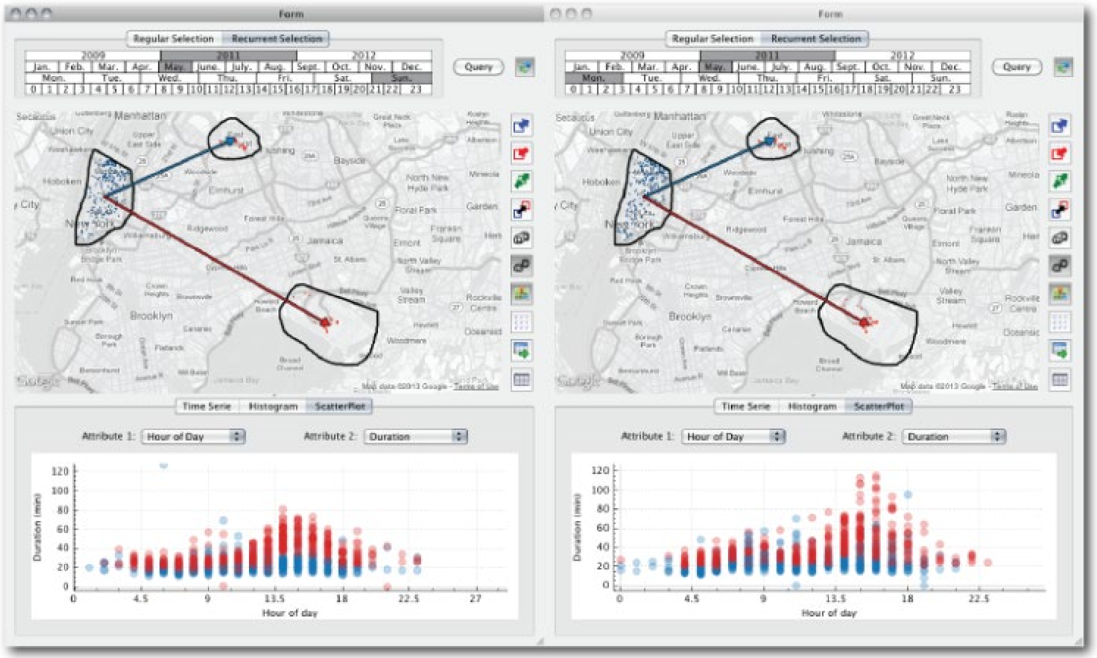
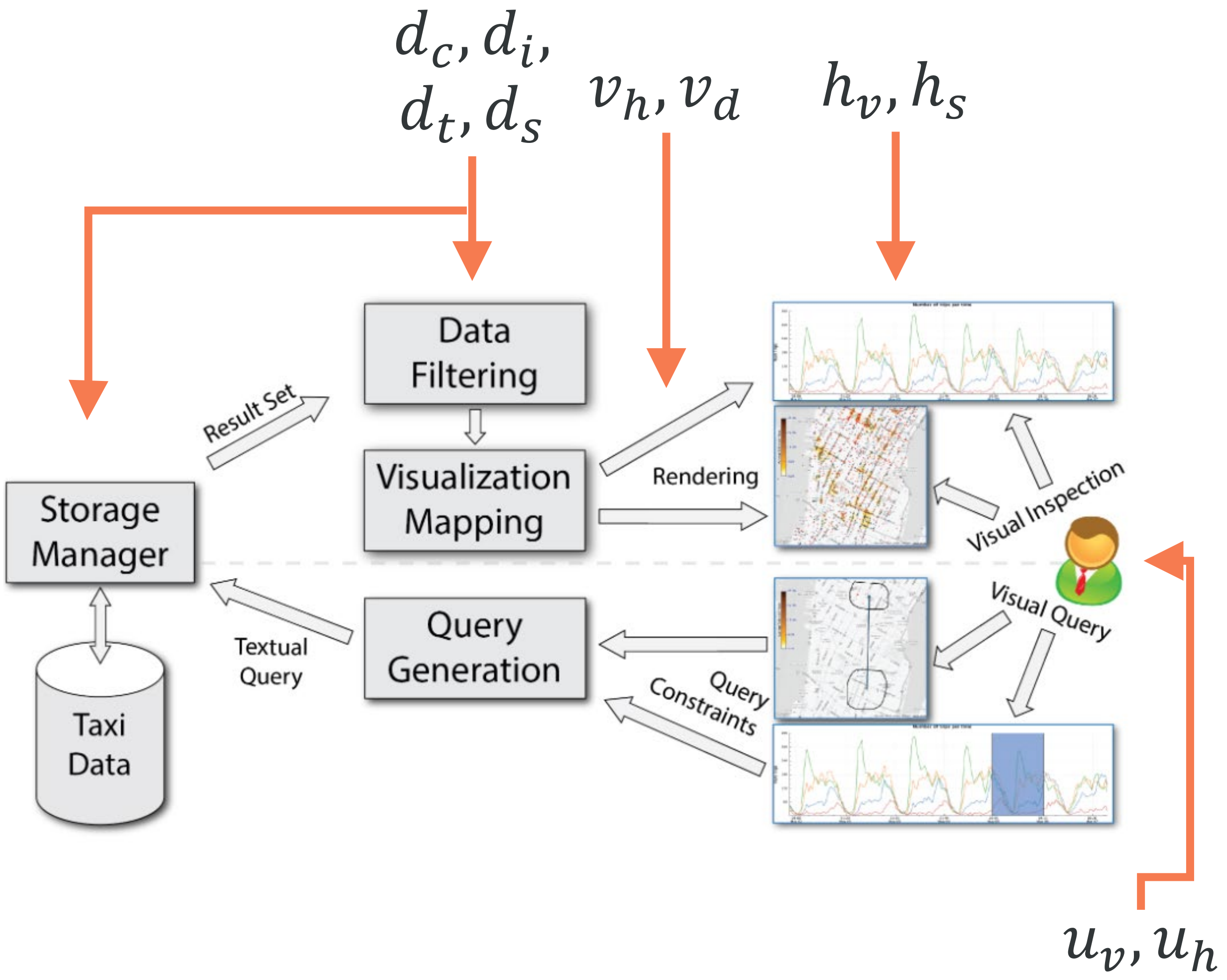
Visual analysis process: TaxiVis





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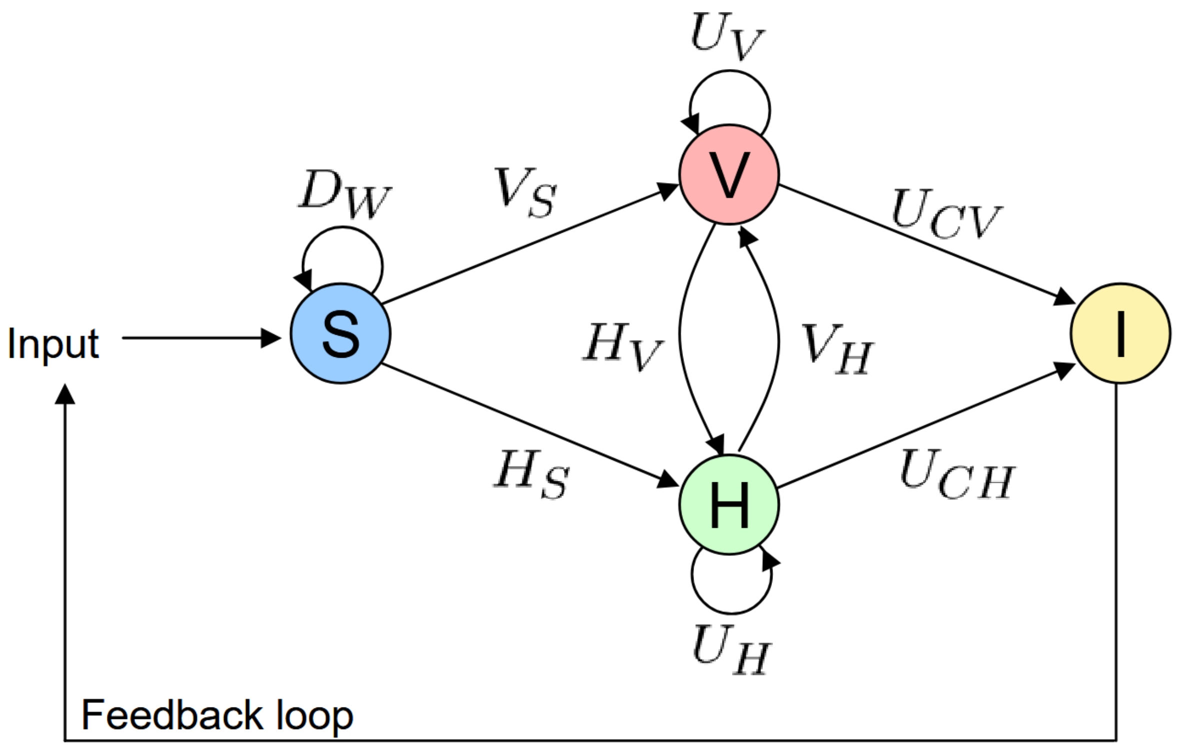
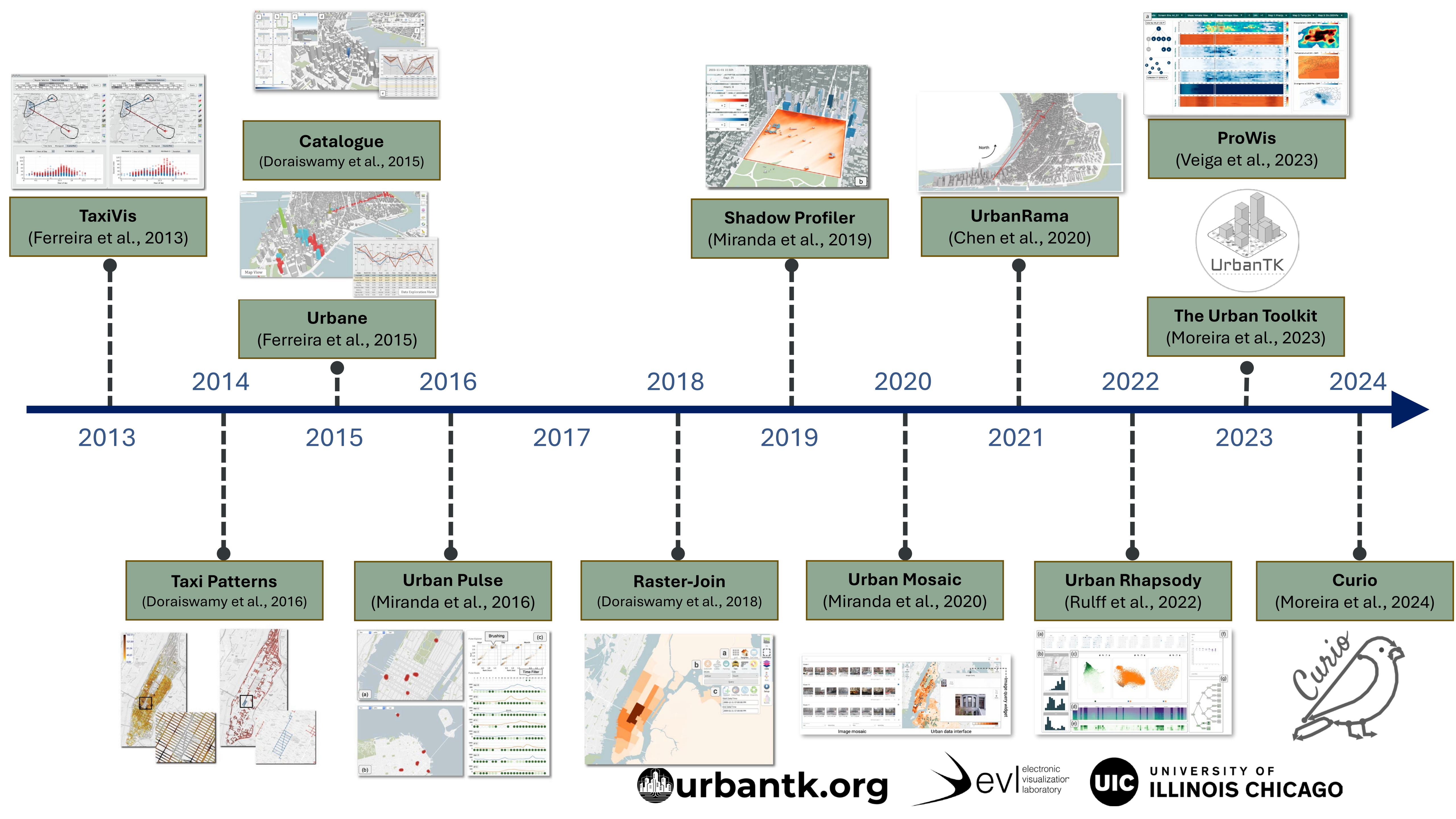


Visual analytics tool that includes:

- h_s: hypotheses using computational topology
- h_v: hypotheses from visualization
- v_s: visualizing data
- v_h: visualizing hypotheses
- u: interaction



TaxiVis
(Ferreira et al., 2013)
Catalogue
(Doraiswamy et al., 2015)
Urbane
(Ferreira et al., 2015)
Shadow Profiler
(Miranda et al., 2019)
UrbanRama
(Chen et al., 2020)
ProWis
(Veiga et al., 2023)
UrbanTK
The Urban Toolkit
(Moreira et al., 2023)
2013
2014
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2016
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2018
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2020
2021
2022
2023
2024
Taxi Patterns
(Doraiswamy et al., 2016)
Urban Pulse
(Miranda et al., 2016)
Raster-Join
(Doraiswamy et al., 2018)
Urban Mosaic
(Miranda et al., 2020)
Urban Rhapsody
(Rulff et al., 2022)
Curio
(Moreira et al., 2024)
Curio
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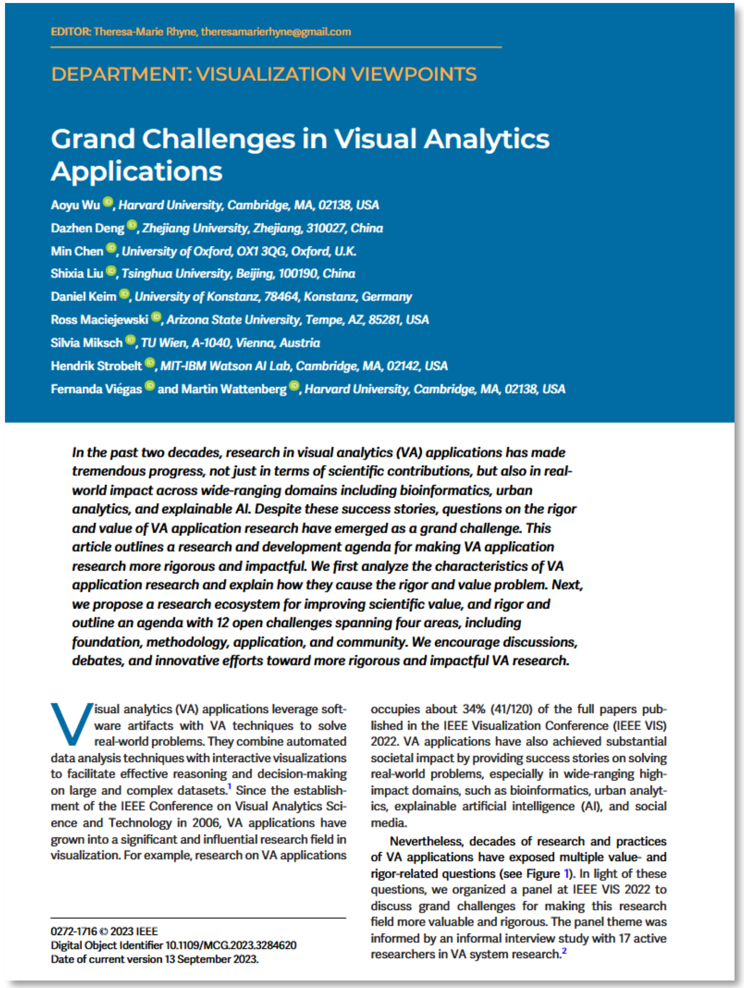
What is a visual analytics system?

What are the practical challenges in VA systems?

What are the research challenges in VA systems?

Grand Challenges in VA applications



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DEPARTMENT: VISUALIZATION VIEWPOINTS

Grand Challenges in Visual Analytics Applications

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Silvia Miksch, TU Wien, A-1040, Vienna, Austria
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In the past two decades, research in visual analytics (VA) applications has made tremendous progress, not just in terms of scientific contributions, but also in real-world impact across wide-ranging domains including bioinformatics, urban analytics, and explainable AI. Despite these success stories, questions on the rigor and value of VA application research have emerged as a grand challenge. This article outlines a research and development agenda for making VA application research more rigorous and impactful. We first analyze the characteristics of VA application research and explain how they cause the rigor and value problem. Next, we propose a research ecosystem for improving scientific value, and rigor and outline an agenda with 12 open challenges spanning four areas, including foundation, methodology, application, and community. We encourage discussions, debates, and innovative efforts toward more rigorous and impactful VA research.

Visual analytics (VA) applications leverage software artifacts with VA techniques to solve real-world problems. They combine automated data analysis techniques with interactive visualizations to facilitate effective reasoning and decision-making on large and complex datasets.[1] Since the establishment of the IEEE Conference on Visual Analytics Science and Technology in 2006, VA applications have grown into a significant and influential research field in visualization. For example, research on VA applications occupies about 34% (41/120) of the full papers published in the IEEE Visualization Conference (IEEE VIS) 2022. VA applications have also achieved substantial societal impact by providing success stories on solving real-world problems, especially in wide-ranging high-impact domains, such as bioinformatics, urban analytics, explainable artificial intelligence (AI), and social media.

Nevertheless, decades of research and practices of VA applications have exposed multiple value- and rigor-related questions (see Figure 1). In light of these questions, we organized a panel at IEEE VIS 2022 to discuss grand challenges for making this research field more valuable and rigorous. The panel theme was informed by an informal interview study with 17 active researchers in VA system research.[2]

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Beyond the more practical challenges, what are the long-term, grand challenges in visual analytics applications?

- Several **value-** and **rigor-**related questions.

Grand Challenges in VA applications

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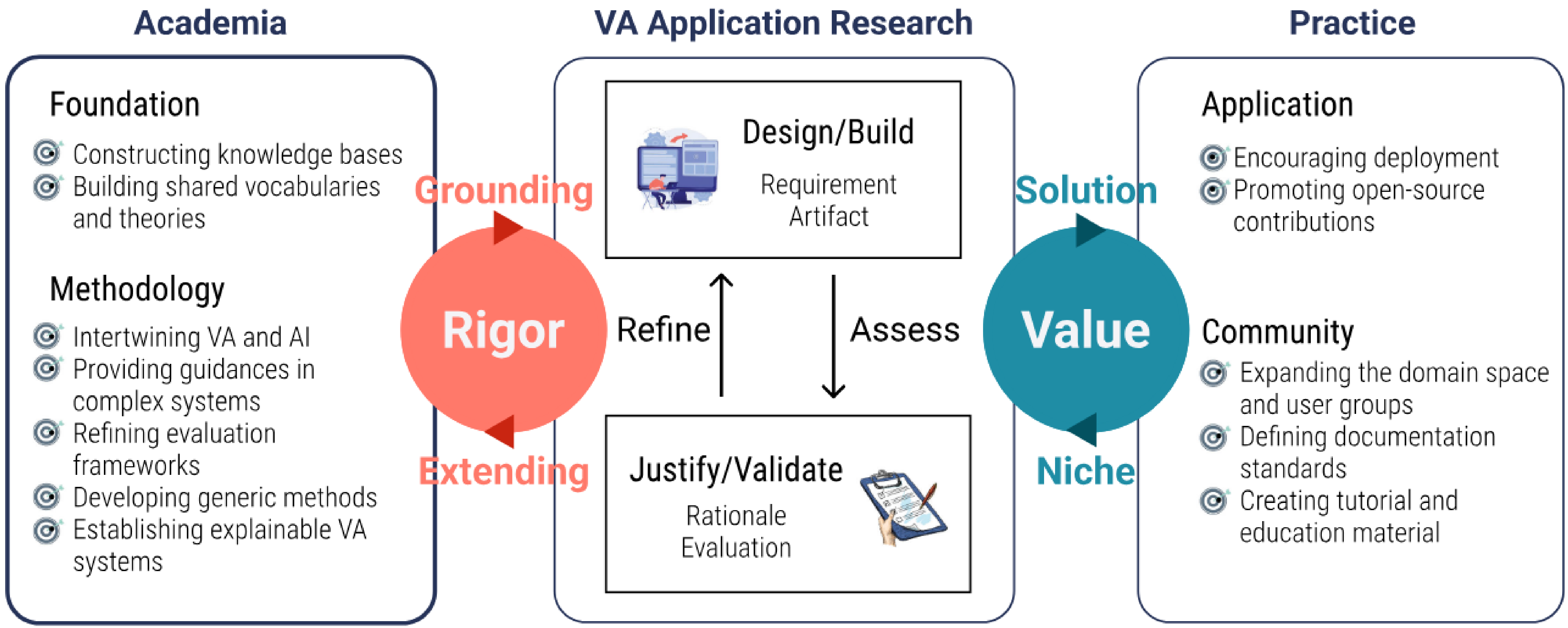
Grand Challenges in VA applications



Grand Challenges in VA applications

Foundation

- **Constructing knowledge bases:** Build a shared, searchable repository of real-world VA "symptoms, causes, remedies, and side effects" to accumulate evidence across applications.
- **Building shared vocabularies and theories:** Establish common terminology and a stronger "theory of VA" that links problem requirements to solution designs in a principled way.

Methodology

- **Intertwining VA and AI:** Make VA and AI mutually strengthen each other, e.g., support data-centric AI and human oversight while improving understanding and bias mitigation.
- **Providing guidance in complex systems:** Offer mixed-initiative assistance (recommendations, cues, alternatives) that is timely, trustworthy, adaptive, and non-disruptive for domain users.
- **Refining evaluation frameworks:** Develop evaluation approaches that fit VA's process-oriented, task/data/user-specific nature (beyond anecdotal case studies), potentially including automated testing.
- **Developing generic methods:** Balance domain-specific solutions with reusable methods and criteria that help transfer VA approaches across tasks, domains, and settings.
- **Establishing explainable VA systems:** Capture and communicate the analytical process (e.g., via provenance) so insights and reasoning can be explained to others and potentially learned from.

Grand Challenges in VA applications

Application

- **Encouraging deployment:** Incentivize field deployment (not just lab prototypes) and create community/review structures that value real-world adoption and longitudinal feedback.
- Promoting open-source contributions: Make systems and components more reusable and sustainable as open-source, including principled modularization instead of monolithic “do-everything” tools.

Community

- Expanding the domain space and user groups: Broaden VA impact by engaging more domains and end users beyond today’s “core clients,” through deeper interdisciplinary collaboration.
- Defining documentation standards: Create standards/guidelines for documenting VA systems (papers, demos, code, workflows) so others can understand, use, and reproduce them.
- Creating tutorial and educational material: Build training materials that cover the full VA application skill set (requirements, implementation, evaluation), not just dashboard construction.

Main takeaways

- General approaches might not scale.
- Big data visual analytics might require structures specifically designed for a given task.
- Understand the data, and how it can be transformed to best fit different computational resources and architectures.
- **At its core, a big data visual analytics system is operationalizing a visual analytics process.**
- **Important grand challenges remain for VA systems.**